

Worksheet 1.2

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1. Aim/Overview of the practical:

Implement a sentimental analysis system using Hugging Face's BERT model

Coding:

```
from transformers import AutoModelForSequenceClassification, AutoTokenizer
import torch
```

```
model_name = "nlpTown/bert-base-multilingual-uncased-sentiment"
tokenizer = AutoTokenizer.from_pretrained(model_name)
model = AutoModelForSequenceClassification.from_pretrained(model_name)
```

```
text = "We should not go out to play today"
```

```
inputs = tokenizer(
    text,
    return_tensors="pt",
    padding=True,
    truncation=True
)
```

```
with torch.no_grad():
    logits = model(**inputs).logits
```

```
probabilities = torch.nn.functional.softmax(logits, dim=-1)
```

```
prediction = torch.argmax(probabilities, dim=1).item()
```

```
sentiment = "Positive" if prediction > 2 else "Negative"
```

```
print("Probabilities:", probabilities)
```

```
print("Predicted label index:", prediction)
```

```
print("Sentiment:", sentiment)
```

OUTPUT:

```
... Probabilities: tensor([[0.3250, 0.2836, 0.2268, 0.0855, 0.0790]])  
Predicted label index: 0  
Sentiment: Negative
```

3. Learning outcomes (What I have learnt):

- Understand the fundamentals of Natural Language Processing (NLP) and sentiment analysis techniques.
- Load and utilize Hugging Face's pre-trained BERT model for sentiment classification tasks.
- Preprocess and tokenize textual data appropriately for transformer-based models.
- Perform sentiment prediction on input text using BERT and interpret the output probabilities.
- Evaluate sentiment analysis results for accuracy, confidence, and real-world applicability.